2320030299-Vandit Manohar
2320030318 Y Pavan Kumar Reddy
2320030335 Sushanth
2320030349 K Sai Prasad

Field Visit Report — FreshTrack

1. Introduction

As part of the development of the *FreshTrack* system, a field visit was conducted to understand real-world practices in inventory management, storage of perishable goods, and food waste handling. The visit aimed to bridge the gap between theoretical system design and practical implementation by observing how grocery items and assets are managed in real environments. Special focus was given to refrigeration systems and supermarket operations, as they play a critical role in preserving perishable items.

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2. Objective of the Visit

The primary objectives of the field visit were:

- To study how supermarkets manage inventory and expiry tracking
- To understand the role of refrigeration in preserving food items
- To observe real-time stock handling and storage practices
- To identify common issues leading to food spoilage and wastage
- To gather requirements for designing an efficient digital solution

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3. Location of Visit

The field visit was conducted at:

- **Appliance Store (Refrigerator Sales and Storage Solutions)**
  A retail store specializing in refrigerators and cooling appliances, where insights were gathered about different types of refrigeration systems, storage capacities, and their importance in extending the shelf life of perishable goods.
- **Supermarket**
  A local supermarket where daily operations such as inventory management, stock rotation, and expiry monitoring were observed.

(You can replace with actual store names if required by your college.)

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4. Methodology

The following methods were used during the visit:

- Direct observation of storage and inventory practices
- Interaction with store staff and supermarket employees
- Understanding refrigeration usage and temperature control

- Analysis of stock arrangement and expiry labeling systems
- Informal discussions about challenges faced in daily operations

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5. Observations

The following key observations were made:

- Supermarkets follow **stock rotation methods** like FIFO (First In, First Out), but not always strictly
- Expiry dates are printed but **not actively tracked using digital systems** in smaller setups
- Refrigerators play a crucial role in preserving dairy, vegetables, and frozen products
- Improper temperature settings can lead to faster spoilage
- Manual monitoring is still widely used, increasing chances of human error
- Overstocking of perishable items sometimes results in wastage
- Appliance stores highlighted that users often **underutilize advanced fridge features** like humidity control

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6. Problems Identified

Based on the observations, the following problems were identified:

- Lack of automated systems for tracking expiry dates
- Dependence on manual inventory management
- Inefficient use of refrigeration technology
- Limited awareness about proper storage practices
- No real-time alerts for expiring or spoiled items
- Difficulty in maintaining optimal stock levels

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7. Relevance to FreshTrack

The field visit directly contributed to refining the features of the *FreshTrack* system:

- **Expiry Tracking System** → Inspired by lack of active monitoring
- **Inventory Management Module** → Designed to reduce overstocking and understocking
- **Notification System** → Addresses absence of alerts in current practices

- **Asset Management** → Includes tracking of appliances like refrigerators
- **User-Friendly Design** → Ensures accessibility for both households and shop owners

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8. Outcome of the Visit

The field visit helped in:

- Understanding real-world storage and inventory challenges
- Identifying gaps in current systems
- Improving system features based on practical needs
- Validating the importance of integrating technology in daily inventory management

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9. Conclusion

The field visit to both the refrigerator appliance store and the supermarket provided valuable insights into the handling, storage, and management of perishable goods. It highlighted the importance of proper refrigeration and efficient inventory tracking in reducing food wastage. The study confirmed that existing systems rely heavily on manual processes, which are prone to errors and inefficiencies. Based on these findings, *FreshTrack* has been designed as a smart, automated solution that enhances inventory visibility, ensures timely notifications, and promotes sustainable consumption practices.

10. Photography proofs:

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